



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Sangam - Your City Your Experience

Team ID: DSCPP-III-2026-T059

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27



Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T059

Semester: 3rd

Section: A , DS1, ML1

Domain: *Computer Science*

Team Members

1. *Srishti Dhasmana – 2510010598*
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Mentor

Mr.Manish Mehta

Interactions completed: 4



Problem Statement & Motivation

Problem Statement

Tourists visiting a new city often struggle to plan their visit efficiently, while most map and review platforms only show scattered lists of places with no real sense of order. This gap exists largely because route planning across multiple stops is a genuinely hard combinatorial problem, and most existing tools sidestep it rather than solve it. Our project aims to bridge this divide by helping users mark, browse, and organize places into a structured, optimized travel plan.

Motivation

- ▶ *No structured way to plan a multi-stop visit.*
- ▶ *Existing apps show places, not usable routes.*
- ▶ *No efficient method to organize and search large place datasets.*

Target Users / Stakeholders

- ▶ *Tourists and first-time visitors to a city.*
- ▶ *Hotels, restaurants, and attractions seeking visibility.*
- ▶ *Local guides and travel planners needing structured tools.*



Objectives

Project Objectives

1. *Develop a web-based solution to simplify city exploration and trip planning.*
2. *Implement core features using C++ and DSA concepts.*
3. *Apply DSA logic (graphs, searching, sorting) to structure routes and place data.*
4. *Test the system for accuracy and performance.*
5. *Demonstrate real-world use and future scope for expansion.*

Key Objective: *"Enable users to mark places on a map and generate an optimized, structured travel plan, with measurable improvement in route efficiency over manual planning."*



Proposed Solution

Proposed Approach

1. *Input / Data Collection*
2. *Processing / Core Algorithm*
3. *System Logic*
4. *Database / Storage*
5. *Output / Visualization*

Key Features

- ▶ *Interactive Map with Place Markers*
- ▶ *Place Search and Autocomplete*
- ▶ *Route and Itinerary Optimization*
- ▶ *Category-based Browsing*
- ▶ *Route Visualization*

An interactive platform is designed to help tourists explore a city and plan visits using C++ and DSA concepts. User inputs such as selected places and preferences are collected and processed through efficient graph and search algorithms. The system logic computes optimized routes and handles edge cases such as unreachable or duplicate places, while structured data storage using vectors, hash tables, and graphs enables quick retrieval and scalability. Finally, a map-based visual output shows markers and computed routes, making trip planning intuitive, accessible, and efficient while reinforcing computational thinking.

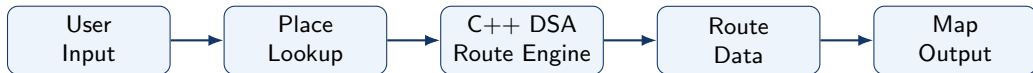


Requirement: Demonstrate meaningful integration of both subjects in **one** project.

Course	Concepts Applied	Project Module
Data Structures in C	<i>Arrays, Linked Lists, Trees, Graphs, Hashing, etc.</i>	Route Engine & Place Lookup Module
OOPs with C++	<i>Classes, Inheritance, Polymorphism, STL, etc.</i>	Core Routing & Itinerary Engine
Web Design	<i>Styling, Navigation, User Interaction, Content</i>	Website Frontend (HTML/CSS/JS + Map Integration)
DBMS	<i>Tables, Normalization, Queries, CRUD Operations</i>	Place & Itinerary Storage Module



System Architecture / Workflow



Major Components

- ▶ Website Frontend (HTML/CSS/JS + Map)
- ▶ Database Module (Place & Itinerary Storage)
- ▶ C++ DSA Engine (Routing & Search)

Data / Control Flow

- ▶ User marks places → website sends selection
- ▶ Engine builds graph → places matched to nodes
- ▶ C++ engine computes route, map displays result to user



Technology Stack & Methodology

Technology Stack

- ▶ Programming: C++, HTML, CSS, JavaScript
- ▶ Data Storage: SQLite (Lightweight DBMS)
- ▶ Tools / Library: Leaflet.js (Map Rendering)
- ▶ IDE: VS Code
- ▶ Version Control: Git / GitHub

Development Methodology

1. Problem & Requirement Analysis
2. Designing Graph Structure & DSA Logic
3. Routing Engine & Website Implementation
4. Route Accuracy Testing
5. Performance & Usability Evaluation

Justify the selection of technologies and methodology.



Initial Progress & Team Contribution

Progress Achieved

- ▶ Background / literature study on existing map & trip-planning tools
- ▶ Requirements identified (place data, route optimization)
- ▶ System architecture & workflow designed
- ▶ Website skeleton (HTML/CSS/JS + map) implemented
- ▶ DSA logic planned for graph structuring & routing

Individual Contribution

Member	Role	Contribution
M1	Frontend & Search Logic	HTML/CSS/JS interface, map integration; implemented trie-based search and hash table lookup in C++
M2	Core DSA & Routing	Graph construction, Dijkstra/A* implementation, itinerary optimization logic in C++
M3	Database & Data Structuring	SQLite schema and persistence layer; structured place data using vectors and linked lists in C++



Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Working city guide platform with interactive map & itinerary planner
- ▶ Functional route optimization across multiple selected places
- ▶ Faster, more efficient trip planning for tourists
- ▶ Scope to expand dataset & add more cities

References

1. E. W. Dijkstra, "A Note on Two Problems in Connexion with Graphs," Numerische Mathematik, 1959
2. P. Hart, N. Nilsson, B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," IEEE Trans. SSC, 1968
3. OpenStreetMap contributors – openstreetmap.org
4. Leaflet.js documentation – leafletjs.com

Thank You
Questions & Suggestions